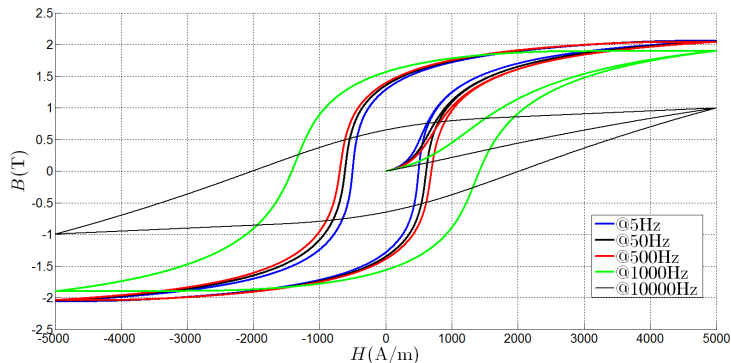


# Para-model control of the Epstein Frame

Preliminary model - the JA magnetic hysteresis model

**(Jiles, Atherton - 1983)**

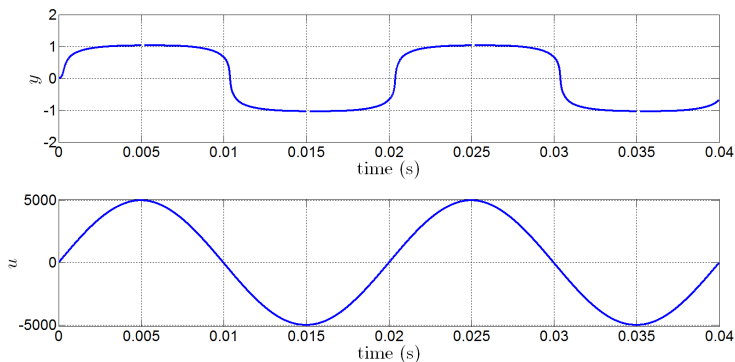
- Control of the JA model as a switched system
- Goal: Drive of  $u \equiv H$  to have  $y \equiv B$  "as close as possible" to a sine (the sine is imposed by the normalized measurements)



# Para-model control of the Epstein Frame

JA magnetic hysteresis model in simulation

- $B$  magnetic response in open-loop according to a  $u$ -sine

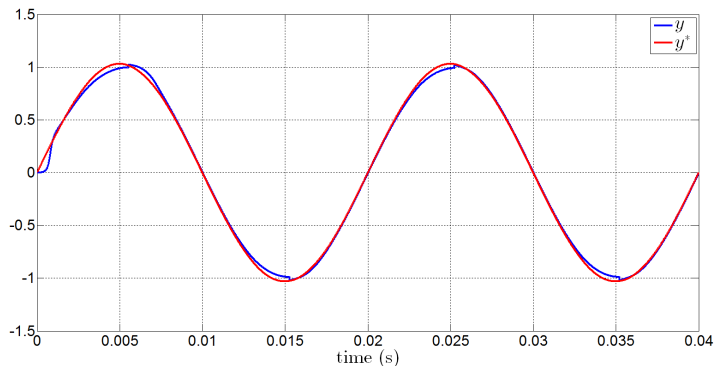


$$M = J_{BH}(H) : \frac{dM}{dH} = \frac{1}{1+c} \frac{M_{an} - M}{\delta k - \alpha(M_{an} - M)} + \frac{c}{1+c} \frac{dM_{an}}{dH}; \quad y = f_{JA}(u) = K_{in}K_e u + K_{out}J_{BH}(K_{in}u).$$

# Para-model control of the Epstein Frame

JA magnetic hysteresis model in simulation

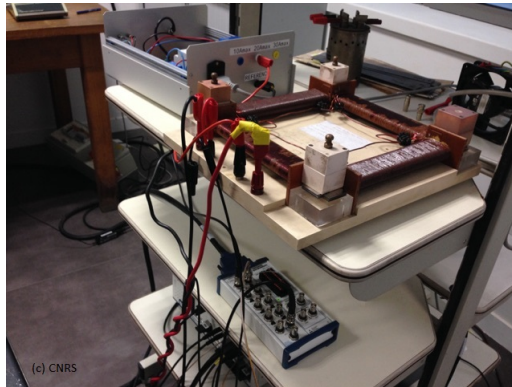
- $B$  magnetic response in closed-loop (switched models)



# Para-model control of the Epstein Frame

## Experimental setup - the Epstein Frame

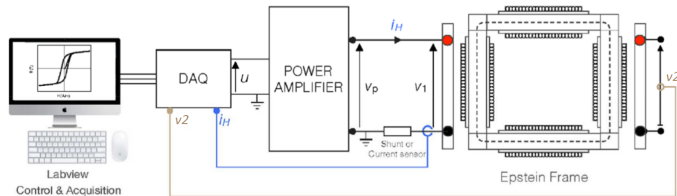
- The JA model is "a part" of the complex Epstein Frame model



# Para-model control of the Epstein Frame

Experimental setup - Synoptic of the practical control procedure

- The HCI and the control are managed by LabVIEW
- A linear power amplifier provides the supply ( $v_1$  voltage)
- Drive of  $v_1$  to control  $v_2$  (to comply with a  $v_2$  reference)

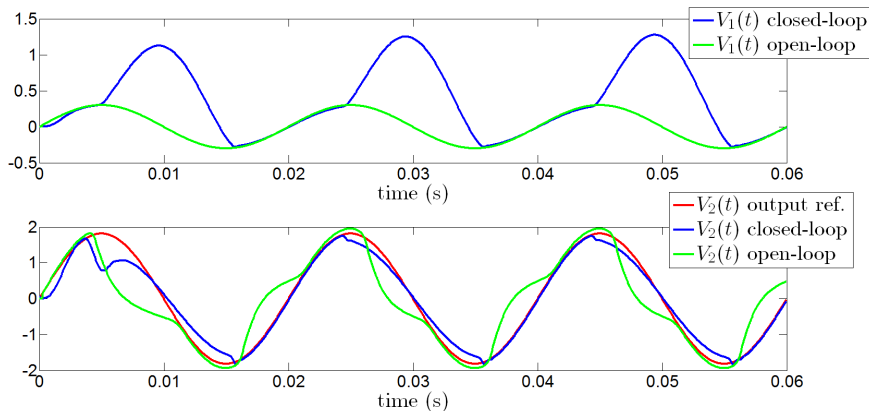


- Development of a full HCI extension to manage the Para-model control

# Para-model control of the Epstein Frame

Experimental results - "sine" control

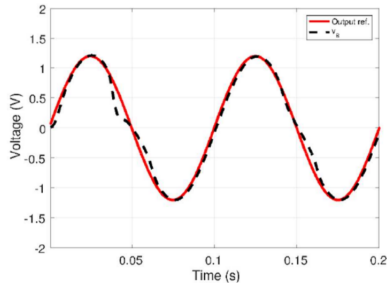
- Open-loop vs closed-loop



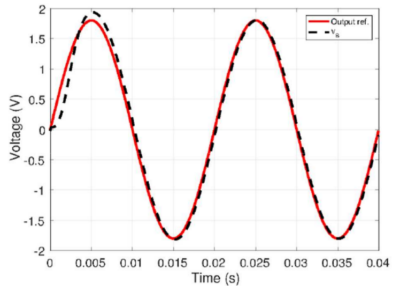
# Para-model control of the Epstein Frame

Experimental results - "sine" control

- Controlled  $v_2$  for a sine reference at 5 Hz and 10 Hz



(a) 5 Hz

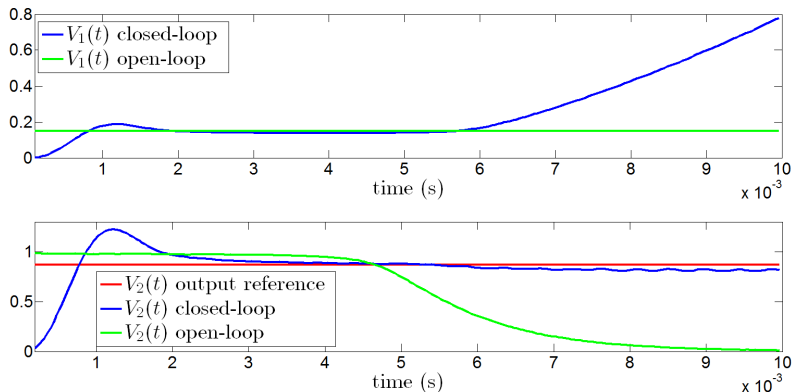


(b) 10 Hz

# Para-model control of the Epstein Frame

Experimental results - "rectangle" control

- Open-loop vs closed-loop

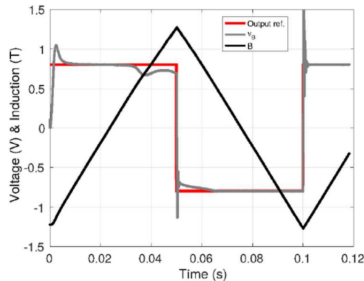




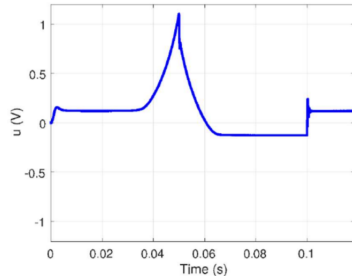
# Para-model control of the Epstein Frame

Experimental results - "rectangle" control

- Controlled  $v_2$  for a rect. reference at 10 Hz with the corresponding calculated  $B$  and  $u$



(a) Controlled  $v_B$  and  $B$



(b)  $u$